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Effect of a Single Aspirin Dose Prior to Fecal Immunochemical Testing on Test Sensitivity for Detecting Advanced Colorectal Neoplasms: A Randomized Clinical Trial

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IMPORTANCE Fecal immunochemical tests for hemoglobin are widely used for colorectal cancer (CRC) screening. Observational studies suggested that sensitivity of fecal immunochemical tests for detecting advanced neoplasms could be increased by acetylsalicylic acid (aspirin), especially among men.

OBJECTIVE To evaluate the potential to increase sensitivity of fecal immunochemical tests by administering a single 300-mg oral aspirin dose 2 days before stool sampling.

DESIGN, SETTING, AND PARTICIPANTS A randomized, placebo-controlled, double-blind trial was conducted in 14 gastroenterology practices and 4 hospitals in Germany, and included 2422 men and women aged 40 to 80 years scheduled for colonoscopy, with no recent use of aspirin or other drugs with antithrombotic effects (enrollment from June 2013 to November 2016, and final follow-up January 27, 2017).

INTERVENTIONS Administration of a single tablet containing 300 mg of aspirin (n = 1208) or placebo (n = 1214) 2 days before fecal sampling for fecal immunochemical test.

MAIN OUTCOME AND MEASURES The primary outcome was sensitivity of a quantitative fecal immunochemical test at 2 predefined cutoffs (10.2 and 17- μ g Hb/g stool) for detecting advanced neoplasms (colorectal cancer or advanced adenoma).

RESULTS Among 2422 randomized patients (mean [SD] age, 59.6 [7.9] years; 1219, 50%, men), 2134 were included in the analysis (78% for primary screening colonoscopy, 22% for diagnostic colonoscopy). Advanced neoplasms were identified in 224 participants (10.5%), including 8 participants (0.4%) with CRC and 216 participants (10.1%) with advanced adenoma. Sensitivity was 40.2% in the aspirin group and 30.4% in the placebo group (difference 9.8%, 95% CI, -3.1% to 22.2%, $P = .14$) at cutoff 10.2- μ g Hb/g stool and 28.6% in the aspirin and 22.5% in the placebo group (difference 6.0%, 95% CI, -5.7% to 17.5%, $P = .32$) at cutoff 17- μ g Hb/g stool.

CONCLUSIONS AND RELEVANCE Among adults aged 40 to 80 years not using aspirin or other antithrombotic medications, administration of a single dose of oral aspirin prior to fecal immunochemical testing, compared with placebo, did not significantly increase test sensitivity for detecting advanced colorectal neoplasms at 2 predefined cutoffs of a quantitative fecal immunochemical test.

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Colorectal cancer (CRC) accounts for approximately 860 000 deaths each year globally.¹ A large proportion of these deaths could be potentially prevented by screening. Randomized clinical trials (RCTs) have demonstrated effectiveness of screening by guaiac-based, chemical fecal occult blood tests (FOBTs) in reducing CRC incidence and mortality.²⁻⁴ Newer fecal immunochemical tests (FITs) for hemoglobin (Hb) outperform guaiac-based FOBTs in diagnostic accuracy⁵⁻⁸ and are broadly recommended for CRC screening.⁹⁻¹¹

FITs detect the majority (70%-80%) of CRCs at high (>90%) specificity but only a minority of advanced adenomas, the precursors of most CRCs.^{8,10} Detecting advanced adenomas in addition to CRCs could make a major contribution to effectiveness of FIT-based CRC screening.¹² Improved sensitivity without simultaneously decreasing specificity could improve the performance of FIT for colorectal cancer screening.

In a previous observational study, enhanced sensitivity of FITs for detecting advanced adenomas was found among participants of screening colonoscopy who used low-dose aspirin for cardiovascular disease prevention.¹³ A plausible explanation might be that aspirin predisposes to subclinical bleeding and, hence, increased detection of advanced adenomas by FIT. This suggests that administration of aspirin prior to fecal sampling might be a practical intervention to increase FIT sensitivity. The objective of this randomized clinical trial was to investigate the effect of a single 300-mg dose of aspirin before FIT on the sensitivity of FITs to detect advanced colorectal neoplasms.

Methods

Ethics Approval

The trial was approved by the Ethics Committee of the Medical Faculty of Heidelberg University, other responsible ethics committees of state physicians' boards, and the German Federal Institute for Drugs and Medical Devices (BfArM). Written informed consent was obtained from each participant.¹⁴

Trial Design and Trial Population

The study protocol and statistical analysis plan are provided in [Supplement 1](#) and [Supplement 2](#), respectively. Briefly, this randomized, double-blind, placebo-controlled trial was designed to evaluate diagnostic performance of 2 different FITs for detecting advanced neoplasms, after a single 300-mg dose of oral acetylsalicylic acid (aspirin) dose compared with placebo. Advanced neoplasms were defined as the presence of either CRC or advanced adenoma that either was 1 cm or larger in size, had tubulovillous or villous components, or high-grade dysplasia. Men and women aged 40 to 80 years with no recent use of aspirin or other drugs with antithrombotic effects were recruited from June 2013 until November 2016 when visiting 1 of 18 trial centers including 14 gastroenterology private practices and 4 hospitals in Germany for a precolonoscopy appointment. In addition to the participants who had been scheduled for primary screening colonoscopy, participants undergoing diagnostic colonoscopy (including colonoscopy conducted before the recommended screening age, which is reimbursed as diagnostic colonoscopy in the German health care

Key Points

Question What is the effect of a single oral dose of aspirin prior to fecal immunochemical testing on test sensitivity for detecting advanced colorectal neoplasms?

Findings In this randomized clinical trial involving 2422 adults aged 40 to 80 years not using aspirin or other antithrombotic medications, there was no statistically significant difference in fecal immunochemical test sensitivity for detecting advanced colorectal neoplasms with administration of a single tablet of 300-mg aspirin, compared with placebo, 2 days before fecal testing (sensitivity at 2 prespecified positivity thresholds: 40.2% vs 30.4%, and 28.6% vs 22.5%, respectively).

Meaning Administration of a single dose of oral aspirin prior to fecal immunochemical testing, compared with placebo, did not significantly increase test sensitivity for detecting advanced colorectal neoplasms at 2 predefined cutoffs of a quantitative fecal immunochemical test.

system) could be included. Participants were excluded if they had previous CRC, had overt or FOBT-detected rectal bleeding, or reported recent use of aspirin or other nonsteroidal anti-inflammatory agents (for details see [eTable 1](#) in [Supplement 3](#)). In each case, the colonoscopy was planned before recruitment to the trial.

Potentially eligible participants obtained detailed written information on the trial from their physicians. Participants received trial medication (allocated in a blinded manner for both physicians and participants according to randomization list), questionnaire, and patient diary. In addition, participants received 4 stool collection kits for each of 2 FITs: FOB Gold Tube Screen (Sentinel Diagnostics), a widely used quantitative FIT,^{8,15,16} and the FD Hb/Hp Complex quick (Frost Diagnostika), a qualitative chromatographic FIT,¹⁷ referred to as quantitative and qualitative FIT hereafter. At the time this trial was planned, both quantitative and qualitative FITs were widely used in Germany.

Participants were asked to collect a baseline stool sample, then take the trial medication and collect further stool samples 2, 3, and 4 days after trial medication intake (1 stool sample per scheduled day, 4 stool samples overall). If stool collection on these days was not possible, postponement to subsequent day(s) was allowed, but all fecal samples were to be collected prior to initiation of large bowel preparation for colonoscopy, which had to be conducted within 3 months after recruitment. Participants were asked to document the day of trial medication intake and the days of stool sample collection in a diary and to fill out a standardized questionnaire addressing sociodemographic characteristics, comorbidities, lifestyle factors, and other potential CRC risk factors. Stool samples, diaries, and questionnaires were sent by regular mail to the coordinating center at the German Cancer Research Center using prepaid and addressed mailing devices.

Randomization

The Pharmacy of the University Heidelberg compiled randomization lists based on computer-generated random numbers (Randomization in Treatment Arms, RITA, version 1.24) and

packed the trial medication (single tablet containing 300-mg aspirin without enteric coating [ASS Ratiopharm 300 mg, Ratiopharm GmbH] or cellulose, lactosemonohydrate, magnesium stearate [placebo]). A randomized block design was used with each block containing 20 tablets (10 assigned to each group). Tablets were taken from their original packaging by trained personnel in a clean room class D environment and repackaged and labeled in a blind manner.

Data Processing

Data collected at the trial centers were entered into electronic case report forms (eCRFs) on site and transferred to a database at the coordinating center. Deidentified colonoscopy and histology reports were sent to the coordinating center, where information was extracted and entered into a colonoscopy database by 2 independent, blinded reviewers. Comprehensive plausibility checks of data entries were performed by medical information specialists.

Laboratory Methods

Participants were asked to collect fecal samples in sampling tubes provided by the manufacturers and send them to the coordinating center, where they were stored at 2°C to 8°C until analysis. Median time between stool sampling and arrival in the coordinating center was 4 days (interquartile range, 3-5 days).

The quantitative FIT was analyzed in an external accredited laboratory (Limbach Laboratory) using Abbott Architect c8000 (analytical range, 0.03-142 µg-Hb/g feces). Analyses of Hb concentration by the qualitative FIT were performed at the coordinating center. All laboratory analyses were performed blinded with respect to treatment assignment and clinical and colonoscopy data.

Outcomes

The primary outcome was sensitivity of the quantitative FIT for detecting advanced neoplasms at 2 predefined cutoffs: 17-µg Hb/g feces, the cutoff given by the manufacturer, and 10.2-µg Hb/g feces, the cutoff of the qualitative FIT, in fecal samples scheduled 2 days after intake of aspirin or placebo. The main analysis followed an intention-to-screen approach using fecal samples scheduled for day 2 after tablet intake; results of analyses following a prespecified per-protocol approach using fecal samples collected at exactly day 2 are additionally reported. Prespecified secondary outcomes reported in this article were specificity, positive predictive value (PPV) and negative predictive value (NPV) of the quantitative test, potential gain in sensitivity by combining results from multiple fecal samples scheduled on any day between day 2 and day 4, sensitivity, specificity, PPV and NPV of the qualitative test, sex-specific sensitivity, specificity, PPV and NPV of both tests, and serious adverse events. Further secondary outcomes not reported in this article were likelihood ratios and area under the curve.

Sample Size and Power Calculations

Assuming a prevalence of advanced neoplasms of 10% and based on the results of the preceding observational study,¹³ the

trial had been designed to have 90% power (in 2-sided testing at $\alpha = .05$) to detect a difference in sensitivity between intervention and control group at day 2 of 24 percentage points (60% vs 36%).

Statistical Analyses

Participants were analyzed according to their randomization group. Those who withdrew their consent or did not contribute with any samples and information were excluded as dropouts. Participants who did not provide stool samples prior to colonoscopy were excluded from the primary analysis.

All analyses were performed with R statistical software version 3.4.0. Wald-adjusted confidence intervals were computed for sensitivity and specificity values (Agresti-Coull adjustment)¹⁸ and the differences (Agresti-Caffo adjustment)¹⁹ with the PropCIs package. Interaction *P* values were obtained from a generalized linear model fit for binomial data. Statistical significance was defined by a 2-sided *P* value < .05. No adjustments were made for multiple testing. Analyses of secondary outcomes should therefore be regarded as exploratory.

Sensitivity Analysis

To account for study conduction at multiple sites, a generalized linear mixed model, with a random intercept accounting for the site, was fitted for each outcome of the main intention-to-screen analysis via the lme4 package.²⁰ A likelihood ratio test²¹ for the null hypothesis of zero variance of the random intercept was performed.

Results

From 2422 recruited participants, 288 dropped out or were excluded for the reasons shown in the **Figure**, leaving 2134 participants for the analysis. **Table 1** displays main characteristics of the trial population, which included similar proportions of women and men. The majority (93.6%) of participants were between 45 and 74 years old, mean age was 59.6 years. Seventy-eight percent of colonoscopies were conducted for primary screening. Advanced neoplasms were identified in 224 participants (10.5%), including 8 participants (0.4%) with CRC and 216 participants (10.1%) with advanced adenoma. Distributions of demographic and clinical characteristics were similar in the intervention and the control group.

eTable 2 in **Supplement 3** shows the numbers of returned and valid FITs by both scheduled and actual day of stool sampling. A total of 1030 (95.8%) and 1012 (95.6%) valid samples for the quantitative FIT scheduled for day 2 were obtained from participants in the intervention and control group, respectively. Numbers for the qualitative FIT were 974 (90.6%) and 969 (91.5%), respectively.

Primary End Point

The sensitivities of the quantitative FIT for detecting advanced neoplasms using fecal samples scheduled 2 days after tablet intake (intention-to-screen analysis) are shown in **Table 2**. At the 10.2-µg Hb/g stool cutoff, sensitivity was 40.2% in the intervention group and was 30.4% in the placebo group

(difference, 9.8%; 95% CI, -3.1% to 22.2%; $P = .14$); at the 17- μ g Hb/g stool cutoff, the sensitivities were 28.6% and 22.5%, respectively (difference, 6.0%; 95% CI, -5.7% to 17.5%, $P = .32$).

Secondary End Points

At the 10.2- μ g Hb/g stool cutoff, specificity of the quantitative FIT was 82.2% in the intervention group and 88.7% in the control group (difference, -6.4%, 95% CI, -9.6% to -3.2%, $P < .001$) and at 17- μ g Hb/g stool cutoff was 91.7% and 94.8%, respectively (difference, -3.1%; 95% CI, -5.4% to -0.8%; $P = .008$) (Table 2).

For the qualitative test, sensitivity was higher by 12.7 percentage points in the intervention group (34.7% vs 22.0%) than it was in the control group (95% CI, 0.1 to 24.7, $P = .048$) but specificity was lower by 8.0 percentage points in the intervention group (83.8% vs 91.8%) than it was in the control group (95% CI, -11.0 to -4.9; $P < .001$).

Among the subgroup of men, point estimates of sensitivity were between 8.4 (95% CI, -7.5 to 23.3) and 17.6 (95% CI, 1.3 to 32.6) percentage points higher, and specificities were between 2.5 (95% CI, -6.1 to 1.1) and 6.6 (95% CI, -11.2 to -2.0) percentage points lower in the intervention group than in the control group (Table 2). In contrast, among women sensitivity was similar but specificity was lower in the intervention group than in the control group (Table 2). P values for interaction effects between sex and intervention were all statistically nonsignificant and ranged from .08 to .49.

eTable 3 in Supplement 3 shows sensitivities and specificities obtained when test positivity was defined by at least 1 positive result in up to 3 fecal samples scheduled for days 2, 3, and 4. Compared with results for day-2 samples only, higher sensitivities and lower specificities were observed in both the intervention group and the control group, with little change in the differences in sensitivity between the intervention and control groups.

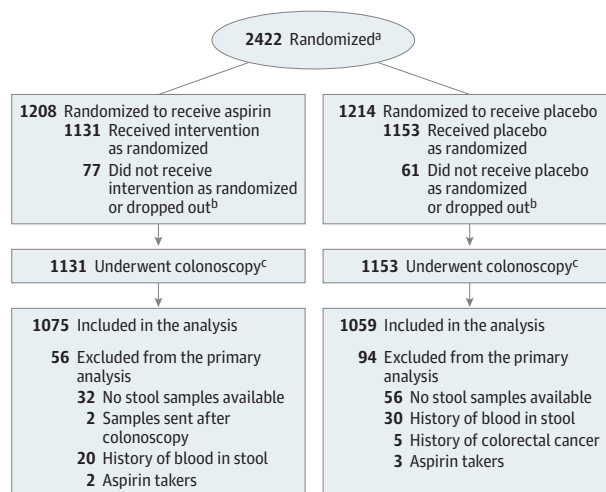
eTable 4 in Supplement 3 shows sensitivities and specificities using fecal samples collected exactly 2 days after tablet intake (per-protocol analysis). Compared with intention-to-screen analysis, differences in sensitivity between the intervention and control group were of greater magnitude among men, with sensitivities that were 23.2 (95% CI, 5.0 to 39.5) and 22.5 (95% CI, 4.5 to 38.6) percentage points higher in the intervention group at the lower cutoff of the quantitative test and for the qualitative test, respectively. Differences in specificity between the intervention and control groups were similar for the per-protocol and intention-to-screen analyses.

Table 3 and eTable 5 in Supplement 3 provide PPVs and NPVs at day 2 after participants took a single 300-mg aspirin dose or placebo. Overall, no significant differences were seen, neither in intention-to-screen nor in per-protocol analysis.

Adverse Events

Overall, the trial medication was well tolerated. In the aspirin group, 11 adverse events in 7 participants were reported. Two serious adverse events in 2 participants (acute appendicitis and acute suppurative cholangitis accompanied by acute kidney injury), which required hospitalization, were considered as not

Figure. Flow Diagram of Recruitment and Exclusions



^a The number of participants assessed for eligibility was not collected.

^b Patients who did not contribute with any sample or information were considered as dropouts.

^c Patients who completed the colonoscopy were considered to have been followed up.

Table 1. Characteristics of the Trial Population

	Group, No. (%)	
	Aspirin (n = 1075)	Placebo (n = 1059)
Sex		
Men	547 (50.9)	528 (49.9)
Women	528 (49.1)	531 (50.1)
Age, y		
<45	26 (2.4)	30 (2.8)
45-49	63 (5.9)	52 (4.9)
50-54	165 (15.3)	164 (15.5)
55-59	315 (29.3)	318 (30.0)
60-64	232 (21.6)	216 (20.4)
65-69	144 (13.4)	153 (14.4)
70-74	92 (8.6)	83 (7.8)
≥75	38 (3.5)	43 (4.1)
Primary indication for colonoscopy ^a		
Screening	805 (77.9)	794 (78.2)
Diagnostic	228 (22.1)	221 (21.8)
Most advanced finding at colonoscopy		
Colorectal cancer	3 (0.3)	5 (0.5)
Advanced adenoma	112 (10.4)	104 (9.8)
Nonadvanced adenoma	286 (26.6)	280 (26.4)
None of above	674 (62.7)	670 (63.3)

^a Information on primary indication for colonoscopy was missing for 42 participants in the aspirin group and 44 participants in the control group

related to the trial drug. In the placebo group, 5 participants reported 6 adverse events. Serious adverse events were not reported. All participants recovered completely.

Table 2. Sensitivity and Specificity for Detecting Advanced Neoplasms at Day 2 After a Single 300-mg Aspirin Dose or Placebo^a

Test	Cutoff, μg , Hb/g	No. of Detections												Difference, % (95% CI)	
		Aspirin						Placebo							
		TP	FN	TN	FP	Sensitivity, %	Specificity, %	TP	FN	TN	FP	Sensitivity, %	Specificity, %	Sensitivity	Specificity
All participants															
Quantitative	10.2	45	67	755	163	40.2	82.2	31	71	807	103	30.4	88.7	9.8 (−3.1 to 22.2)	−6.4 (−9.6 to −3.2)
Quantitative	17.0	32	80	842	76	28.6	91.7	23	79	863	47	22.5	94.8	6.0 (−5.7 to 17.5)	−3.1 (−5.4 to −0.8)
Qualitative	10.2	35	66	732	141	34.7	83.8	22	78	798	71	22.0	91.8	12.7 (0.1 to 24.7)	−8.0 (−11.0 to −4.9)
Men															
Quantitative	10.2	33	44	360	90	42.9	80.0	15	39	387	68	27.8	85.1	15.1 (−1.6 to 30.6)	−5.1 (−10.0 to −0.1)
Quantitative	17.0	25	52	408	42	32.5	90.7	13	41	424	31	24.1	93.2	8.4 (−7.5 to 23.3)	−2.5 (−6.1 to 1.1)
Qualitative	10.2	28	43	354	74	39.4	82.7	12	43	400	48	21.8	89.3	17.6 (1.3 to 32.6)	−6.6 (−11.2 to −2.0)
Women															
Quantitative	10.2	12	23	395	73	34.3	84.4	16	32	420	35	33.3	92.3	1.0 (−19.1 to 21.4)	−7.9 (−12.0 to −3.8)
Quantitative	17.0	7	28	434	34	20.0	92.7	10	38	439	16	20.8	96.5	−0.8 (−17.9 to 17.2)	−3.7 (−6.7 to −0.8)
Qualitative	10.2	7	23	378	67	23.3	84.9	10	35	398	23	22.2	94.5	1.1 (−17.7 to 20.9)	−9.6 (−13.5 to −5.5)

Abbreviations: FN, false-negative; FP, false-positive; Hb, hemoglobin; TN, true-negative; TP, true-positive.

^a Results of the intention-to-screen analysis.Table 3. Positive and Negative Predictive Values for Detecting Advanced Neoplasms at Day 2 After a Single 300-mg Aspirin Dose or Placebo^a

Test	Cutoff μg Hb/g	Aspirin		Placebo		Difference (95% CI)	
		PPV	NPV	PPV	NPV	PPV	NPV
All participants							
Quantitative	10.2	21.6	91.8	23.1	91.9	-1.5 (-10.7 to 7.4)	-0.1 (-2.7 to 2.5)
Quantitative	17.0	29.6	91.3	32.9	91.6	-3.2 (-17.2 to 10.5)	-0.3 (-2.8 to 2.3)
Qualitative	10.2	19.9	91.7	23.7	91.1	-3.8 (-14.4 to 6.5)	0.6 (-2.1 to 3.3)
Men							
Quantitative	10.2	26.8	89.1	18.1	90.8	8.8 (-3.0 to 19.8)	-1.7 (-5.9 to 2.4)
Quantitative	17.0	37.3	88.7	29.5	91.2	7.8 (-10.3 to 24.8)	-2.5 (-6.4 to 1.4)
	10.2	27.5	89.2	20	90.3	7.5 (-6.4 to 20.2)	-1.1 (-5.3 to 3.0)
Women							
Quantitative	10.2	14.1	94.5	31.4	92.9	-17.3 (-31.8 to -2.5)	1.6 (-1.7 to 4.8)
Quantitative	17.0	17.1	93.9	38.5	92	-21.4 (-42.2 to 0.8)	1.9 (-1.4 to 5.2)
Qualitative	10.2	9.5	94.3	30.3	91.9	-20.8 (-37.8 to -4.0)	2.3 (-1.1 to 5.8)

Abbreviations: Hb, hemoglobin; NPV, negative predictive value; PPV, positive predictive value.

^a Results of the intention-to-screen analysis.

Sensitivity Analysis

Results of the sensitivity analysis for the effect of study site are shown in eTable 6 in Supplement 3. The likelihood ratio test for the null hypothesis of zero variance of the random intercept never achieved statistical significance (P values 0.16 to 0.50), therefore the model without random intercept was retained.

Discussion

In this RCT involving adults aged 40 to 80 years not using aspirin or other antithrombotic medications, administration of

a single dose of oral aspirin prior to fecal immunochemical testing, compared with placebo, did not result in a statistically significant increase in test sensitivity for detecting advanced colorectal neoplasms at 2 predefined cutoffs of a quantitative fecal immunochemical test.

The lack of statistically significant findings in this study contrast with findings of a previous observational study involving similar populations that motivated the conduct of this trial.¹³ The observational study found that the sensitivity of 2 other FITs (RIDASCREEN Haemo-/Haptoglobin Complex) for detecting advanced neoplasms was markedly increased (at a modest decrease in specificity) in users of low-dose aspirin among men but not among women. However, this study

compared regular daily use of low-dose aspirin with no use in a nonrandomized design. An observational study from Israel also found substantially higher sensitivity of yet another FIT (OC-Micro) among users of aspirin or nonsteroidal anti-inflammatory drugs.²² Taken together, results from these observational studies and this trial suggest that further research is needed to assess whether low-dose aspirin may have an effect on FIT test performance, such as in a larger well-powered trial. This trial was designed to detect a 24% absolute increase in sensitivity and was not adequately powered to detect small differences that may nevertheless be clinically meaningful given the low morbidity observed, the low cost of a single dose of aspirin, and the ease of implementation of this intervention across health systems.

Antithrombotic effects of aspirin are well established.²³ They seem to be stronger among men than women,²⁴⁻²⁶ and sex differences in platelet number, function and responsiveness to aspirin have been reported.²⁶⁻²⁸ Also, longer colonic transit time of feces and higher prevalence of constipation among women^{29,30} might favor intracolonic Hb degradation, especially in the case of bleeding from proximal neoplasms that are relatively more common among women.³¹ By contrast, prevalence of conditions potentially associated with false-positive FITs, especially under aspirin treatment, such as hemorrhoids, seems not to show major sex differences.³² Additional adequately powered studies are needed to evaluate potential sex differences and reasons for differential aspirin effects on FIT sensitivity.

Other important questions include necessary aspirin dose and preparation and optimal timing of aspirin intake and fecal sampling. This trial tested a 300-mg dose to achieve a relevant hemorrhagic effect even with a single tablet. Possibly, an equivalent or better effect might be achieved with single or multiple tablets (taken over several days) containing lower doses of aspirin, such as 75 mg or 100 mg, which should be explored in further research. FIT performance 2 days following the dose of aspirin was chosen as the primary analysis to allow for adequate colonic transit time. Sensitivity analyses do not suggest performing fecal sampling over multiple days would improve diagnostic performance of FIT.

Limitations

This trial has several limitations. First, despite the overall large number of participants undergoing FIT screening with or without aspirin, estimates of sensitivity were based on relatively few participants with advanced neoplasms. The trial was designed to detect a large effect size of 24% absolute increase in sensitivity of FIT and was not designed to detect more moderate but nevertheless clinically meaningful between-group differences. Second, multiple secondary analyses were conducted without adjustments for multiple testing. Third, although the sample size for analyses of sensitivity was mainly limited by the low prevalence of advanced neoplasms, sample sizes for the primary analyses were reduced by approximately 4% due to exclusions. However, such exclusions were unrelated to treatment group and should not have biased between-group comparisons. Fourth, the limited size of the study sample precluded more detailed analyses of sensitivity for specific subtypes of advanced neoplasms, eg, by advanced adenoma size. Fifth, the number of potential participants excluded due to pre-existing use of aspirin or nonsteroidal agents was not quantified. Sixth, the trial included participants scheduled for either screening or diagnostic colonoscopies. However, the 22% of the trial participants who had FIT prior to diagnostic colonoscopies were included only if the indication was unrelated to rectal bleeding and therefore was unlikely to have an effect on FIT results. Seventh, only effects on diagnostic performance of 1-time screening were assessed. FIT-based screening programs typically include multiple screening rounds, and potential effects on detection of advanced neoplasms and reduction of CRC incidence and mortality in the long run are yet to be determined.

Conclusions

Among adults aged 40 to 80 years not using aspirin or other antithrombotic medications, administration of a single dose of oral aspirin prior to fecal immunochemical testing, compared with placebo, did not significantly increase test sensitivity for detecting advanced colorectal neoplasms at 2 pre-defined cutoffs of a quantitative fecal immunochemical test.

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Drafting of the manuscript: Brenner.

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Data Sharing Statement: See Supplement 4.

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